

Exercise – 7

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Deadline: 29.06.2026 – 1:59pm

Task(s):

- Navigate to the project titled: „ETCE & IoT Mining Exercises 2026“.
- **The exercise should be visible to you in the folder titled „ETCE-Exercise-Solutions/E07“. The older project will not contain E07.**
- The exercise is given to you in the form of a jupyter notebook.
- Programming language: Python 3.10+
 - **You only need to modify the „solution.ipynb“ file.** More detailed instructions on where you need to insert your code can be found in this file.
 - The final cells in the solution.ipynb file grades your solution.
 - This will give you feedback on your solution, if any of the „assert“s receive an error, that indicates that you need to fix the indicated function(s).
- To submit your solution, simply save your ‚solution.ipynb‘ file.
- Grading will be conducted automatically by cocalc based on the evaluation in the notebook.

Task – IoT Security – Energy sellers

In E05 and E06, you first gathered weather data from different sources (weather sensors/APIs) and aggregated them into a single data set. Subsequently, you processed the data to make predictions based on the results of the gathered information. However, so far, we have discarded the aspect of privacy and security when handling IoT-related data. Especially the transmission of data from the sensors to the processing entity (cloud, edge, etc.) is often prone to data manipulation. Moreover, the sensor data might contain sensitive information that is not meant to be public. Therefore, data in transit must be protected against manipulation and encrypted. In this exercise, the Energy Seller and the Buyer will establish a secure communication channel over a public (“eavesdroppable”) communication channel, by performing an [ECDH \(Elliptic Curve Diffie-Hellman\) Key Exchange](#):

Steps:

1. Use Argon2id and the provided salt to derive a key from the password "ETCE-SS2021-this-is-safe" (without quotes).
2. Load and decrypt the provided private key to be used by your node using the derived key.
3. Load the provided public key of your peer.
4. Generate an ephemeral public and private key pair and sign the public key with your private key. This signed private key shall then be sent to your peer, who will verify it.
5. Receive a signed, ephemeral public key from your peer and verify the signature.
6. Do a key exchange using your ephemeral private key and your peer's ephemeral public key.
7. As an energy seller, use the resulting shared secret to first decrypt an encrypted purchase request from your peer (the energy buyer), compute and encrypt the price for the request (encryption and decryption using an authenticated encryption mechanism) and send it back to your peer.